

COMP3000 COMPUTING PROJECT

2024/2025

PROJECT TITLE

AI BEHAVIOURAL AUTHENTICATION USING MACHINE LEARNING

LINKS

Source code: <https://github.com/Bukka101/AI-behavioural-authentication-by-stephen-nzekwesi>

PROJECT VISION

For secured systems whose aim is to tackle unauthorized access due to improper password management either from the user side or through a data breach, there is a need for proper authentication system which has more features than the traditional username-password method, or even two factor authentication as this tends to be easily bypassed if not implemented correctly.

BhAuth as I call it, is an authentication system that authenticates users not only by username and password, but also by user keystroke dynamics and mouse movement patterns, ensuring first-hand integrity of the user as users' behaviours are unique and cannot be imitated by an intruder.

This system is different from traditional username-password authentication in such a way that it does not only validate the username and password, it also validates the user's keystrokes when filling the login form, and also the mouse movement patterns to spot any difference in behaviour and to verify the integrity of the user.

This could eliminate the need for multi factor authentication through one-time codes or email verifications. As the user behaviour is unique, this would uniquely certify the integrity of a user before granting access.

The project could also extend, if time permits, to cover real time authentication of user behaviour throughout the user session. Continuous authentication helps validate that the user remains the user throughout the session, ensuring first-class integrity of users in a secured system.

This project will employ the benefits of machine learning to learn and recognise user keystroke and mouse behaviours, helping spot an unauthorised user even when the username and password appears to be correct.

RISK PLAN

Considering the project plan and the resources available, there are some risks which are challenging to this project. These are outlined below.

TIME AND CONFLICTING COMMITMENTS

Description: Considering work-life balance as a working student, also bearing in mind the workload of other modules which I must cover, time is a phenomenal factor to this project.

Risk factor: High. Time is an essential factor to the fulfilment of this project. If managed correctly, time and other commitments would not interfere with the progress of this project, but if not, it can hinder the progress and completion of the project, especially considering the deadlines.

Likelihood: Common. There will always be other modules to attend to alongside this project. Work also is continuous and essential for living and settling bills.

Mitigation: to mitigate this, I have set out a strict timetable, with a strict personal punishment for every wasted time. I have also created a clear project backlog and management using Jira. With this, following agile development model to split user stories into smaller and executable tasks, I can be able to work on the project on the go. Also, the bi-weekly supervisor meetings will also help to keep everything on track.

RISK OF FAILURE OF MY DEVELOPMENT PC

Description: My laptop is a bit old, and I am worried that it could develop faults during this project, considering that machine learning requires system resources.

Risk factor: High. My development PC is the basic tool for this project and essentially the most important. Considering how long I have used the PC and system resources required for machine learning, there is a risk of breakdown during the course of this project which can hinder the successful completion of the project.

Likelihood: Not common. Considering that my PC has never developed any fault in my care, the likelihood of this happening is estimated low. It

could probably be that I only overthought the situation that I think it is a challenging risk. Regardless, I have mapped out a proper mitigation plan in an event of such.

Mitigation: To mitigate this, I have optimized my PC to delete heavy and redundant files and programs accumulating resources in the PC for efficient running of the system. I have also set out a sum of money to immediately repair or to acquire a new PC as soon as any breakdown occurs. I have also spoken to a friend who would temporarily lend me his PC in emergency, if the preceding approaches fail to work.

LOSS OF DATA

Description: Data loss has always been a challenging factor for system development over the years. If development data is not properly stored and managed, and data is lost at any stage of this project, it could hinder the progress and completion of this project.

Risk factor: High. Risk of data loss can stampede the progress of this project and can potentially impair the completion of this project considering when, and what data is lost. Noting that the project has a completion deadline.

Likelihood: Common. Data loss is common considering that there are many factors which can contribute to loss of data.

Mitigation: to mitigate this, I have employed the use of a good version control system (GitHub) through this project. This serves a backup and is easily retrievable. Proper commits at every stage will ensure that very little or no data will be lost in an unforeseen circumstance. This will also help a lot in mitigating the risk in an event of my PC break down, as outlined in the above risk.

PRODUCT AND SYSTEM ENVIRONMENTAL RISKS AND SHORTCOMINGS

Description: for the product itself, there are a range of risks that serves as a threat to the proper use and functionality of the product. Risks like SQL injection, man in the middle attack, brute force attack, data leak, password breach, ransomware of the underlying system, and all other risks associated to a web-based system such as this product.

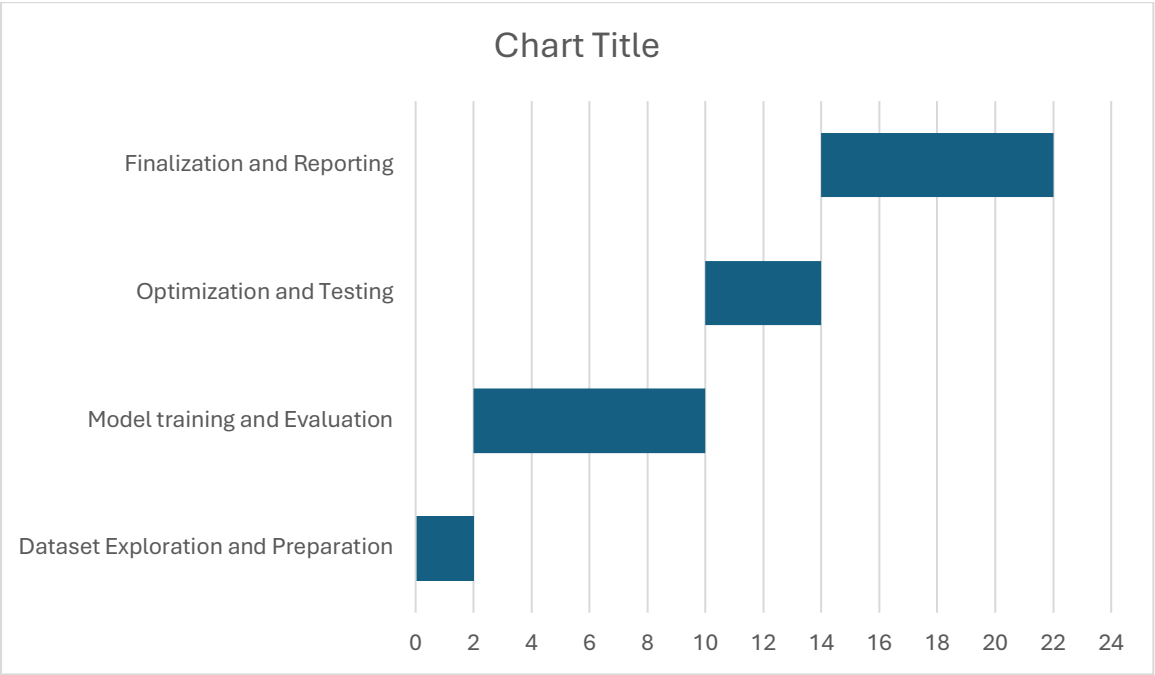
Risk factor: Variable. This depends on the threat itself. It can range from high to low depending on the consequences that the threat in question poses.

Likelihood: Variable. As this is a group factor, the likelihood of individual threats varies. Some like brute force attacks might be common, while some like ransomware might not be as common as the others.

Mitigation: to mitigate this, I will employ security by design principle to mitigate and cut out flaws in the design and development phases of the project. This will help mitigate threats which exploit flaws in design and development. I will also advice further on product best practices at the production stage of this product to help mitigate other web-based threats.

PROPOSED GANTT CHART

Proposed Grantt chart of the project in weeks.



KEYWORDS

Behavioural authentication
Machine learning
AI
Keystroke
Mouse movement